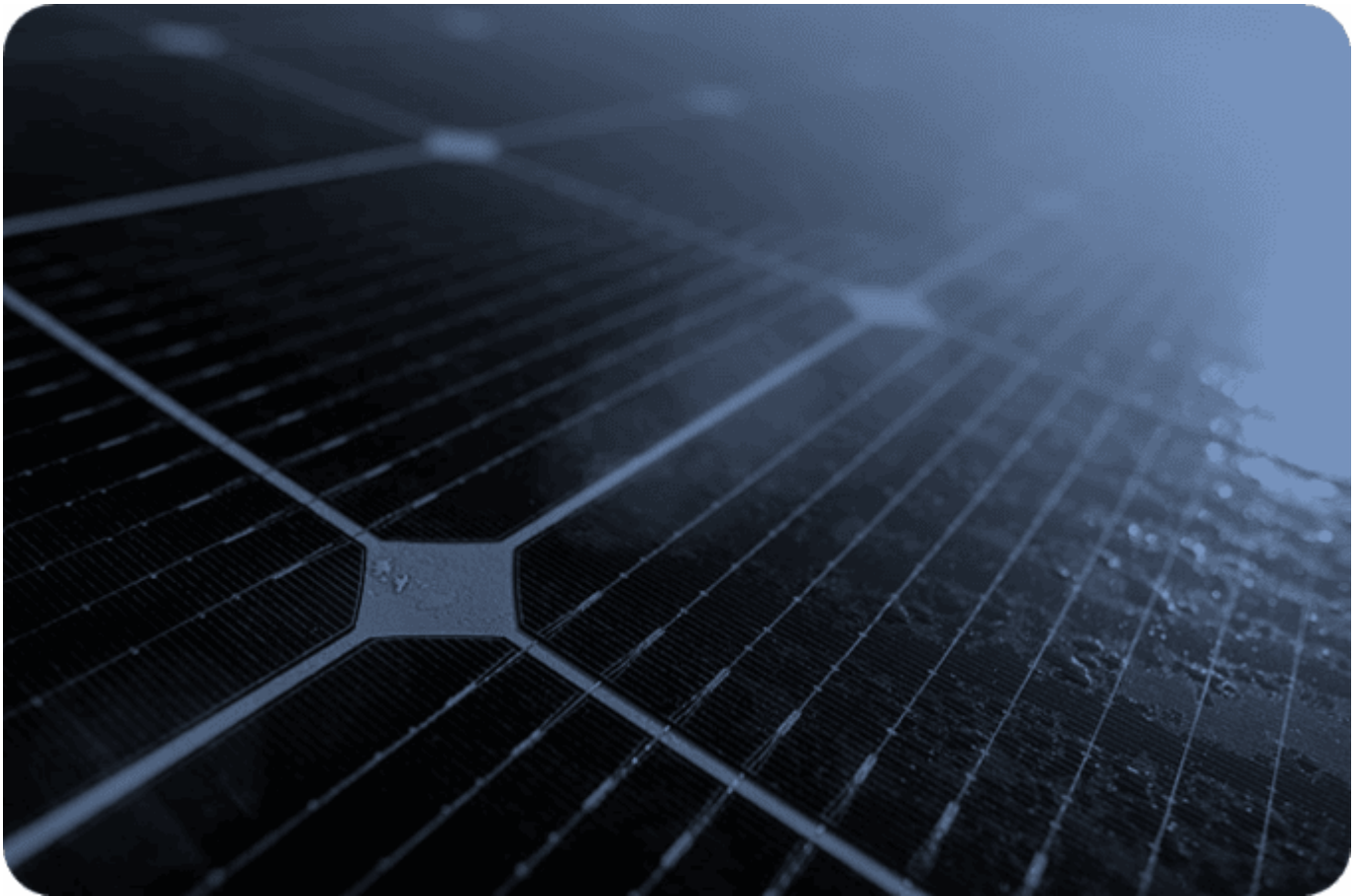


Qualification Pack



Solar PV Cell Manufacturing Technician

QP Code: SGJ/Q0203

Version: 2.0

NSQF Level: 4

Skill Council for Green Jobs || 3rd Floor, CBIP Building, Malcha Marg, Chanakyapuri
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Qualification Pack

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Qualification Pack

SGJ/Q0203: Solar PV Cell Manufacturing Technician

Brief Job Description

Solar PV Cell Manufacturing Technician performs the solar cell manufacturing tasks in a production line, cleans silicon wafer, undertake chemical and thermal processing in making solar cells. The job holder operates various stations including metallization processes and is also responsible for testing and packaging of solar cells.

Personal Attributes

The candidate must be diligent, hardworking, organised and a team player to be successfully able to complete the job safely and timely. She/He must also demonstrate strong work ethics, communicate courteously with co-workers and must be good with following instructions of the supervisor.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SGJ/N3105: Describe the basics of PV Cell, prepare and identify the bill of material and its specification](#)
2. [SGJ/N3104: Perform Solar cell manufacturing- Chemical Process](#)
3. [SGJ/N3103: Perform Solar cell manufacturing-Thermal Process](#)
4. [SGJ/N3102: Perform Solar cell manufacturing- Metallization Process](#)
5. [SGJ/N3101: Perform packing of solar PV Cell](#)
6. [SGJ/N0147: Maintain personal health & safety in a manufacturing facility](#)
7. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Process Line Operator
Country	India

Qualification Pack

NSQF Level	4
Credits	10
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7422.1901 Assembly Line Operator
Minimum Educational Qualification & Experience	12th grade Pass with NA of experience OR Completed 2nd year of the 3-year diploma after 10 with NA of experience OR 10th grade pass plus 2-year NTC with NA of experience OR 10th grade pass (with 1-year NTC plus 1 year NAC) with NA of experience OR 10th grade pass with 3 Years of experience Relevant Experience OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience Relevant Experience OR Previous relevant Qualification of NSQF Level (3.0) with 3 Years of experience Relevant Experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	24/02/2029
NSQC Approval Date	25/02/2026
Version	2.0
Reference code on NQR	QG-04-GJ-05065-2025-V1-SCGJ
NQR Version	1

Remarks:

Total 420 Hours (Theory: 130 hours + Practical: 110 hours + 60 hours of employability skills + 120 hours of OJT)

Qualification Pack

SGJ/N3105: Describe the basics of PV Cell, prepare and identify the bill of material and its specification

Description

This NOS covers the fundamentals of solar PV cells, including basic concepts of solar energy, semiconductor properties, solar cell technology, and identification of bill of materials (BOM) along with their specifications used in solar cell manufacturing.

Scope

The scope covers the following :

- Basics of solar PV cells and solar energy concepts.
- Identification of BOM and material specifications.

Elements and Performance Criteria

Basic of Solar Energy and PV Cell

To be competent, the user/individual on the job must be able to:

- PC1.** Explain the role of Solar Photovoltaic (PV) Cell Manufacturing technician, its importance in the sector and the advantages of doing the course.
- PC2.** Definition of properties of light, Global Horizontal Irradiation (GHI) and Direct Normal Irradiance
- PC3.** Discuss the property of Semiconductor and P-N Junction.
- PC4.** Concept of Solar energy, Working of solar cell, Electrical Properties of Solar Cell, Solar PV Applications
- PC5.** Discuss mono and Poly crystalline Silicon Solar cell technology and recent trends.
- PC6.** Discuss the basic aspects of solar energy and power generation from solar energy
- PC7.** Demonstrate how crystal(mono and Poly crystalline) influence performance of solar cell.
- PC8.** Demonstrate how to interpret signs, notices and/or cautions of shop floor.
- PC9.** Show Solar Cell Manufacturing Process Flow Chart
- PC10.** Show how to provide an overview of Machine Parts, Raw Material to Front Grid (FG) Cell.

Prepare and identify the BOM and its Specification for Solar PV Cell

To be competent, the user/individual on the job must be able to:

- PC11.** Discuss various Materials used in solar cell manufacturing
- PC12.** Silicon Wafer, Chemicals, PPEs, Carrier & Trolley, Gases, Quartz Boat, Graphite Boat Spare, Tweezer, Tube cleaning rod, Printing Screen, Squeegee, Paste, Squeegee holder, Paper Roll, etc.
- PC13.** Technical Specifications & uses, role /importance in cell performance.
- PC14.** Demonstrate and show each component and explain their application in Solar cell
- PC15.** Prepare the BOM of Solar cell

Knowledge and Understanding (KU)

Qualification Pack

The individual on the job needs to know and understand:

- KU1.** Basics of solar energy and working principle of PV cells.
- KU2.** Properties of semiconductors and PN junction in solar cells.
- KU3.** Types of solar cell technologies such as mono and polycrystalline silicon.
- KU4.** Solar PV cell manufacturing process overview.
- KU5.** Materials used in solar cell manufacturing.
- KU6.** Identification of bill of materials (BOM) and their specifications.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Identify and organize materials required for solar cell manufacturing.
- GS2.** Interpret signs, notices, and instructions on the shop floor.
- GS3.** Maintain accuracy while preparing the bill of materials (BOM).
- GS4.** Follow safety and workplace procedures during operations.
- GS5.** Handle materials and tools carefully to avoid damage.
- GS6.** Communicate and coordinate effectively with team members.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Basic of Solar Energy and PV Cell</i>	30	20	-	-
PC1. Explain the role of Solar Photovoltaic (PV) Cell Manufacturing technician, its importance in the sector and the advantages of doing the course.	5	-	-	-
PC2. Definition of properties of light, Global Horizontal Irradiation (GHI) and Direct Normal Irradiance	5	-	-	-
PC3. Discuss the property of Semiconductor and P-N Junction.	5	-	-	-
PC4. Concept of Solar energy, Working of solar cell, Electrical Properties of Solar Cell, Solar PV Applications	5	-	-	-
PC5. Discuss mono and Poly crystalline Silicon Solar cell technology and recent trends.	5	-	-	-
PC6. Discuss the basic aspects of solar energy and power generation from solar energy	5	-	-	-
PC7. Demonstrate how crystal(mono and Poly crystalline) influence performance of solar cell.	-	5	-	-
PC8. Demonstrate how to interpret signs, notices and/or cautions of shop floor.	-	5	-	-
PC9. Show Solar Cell Manufacturing Process Flow Chart	-	5	-	-
PC10. Show how to provide an overview of Machine Parts, Raw Material to Front Grid (FG) Cell.	-	5	-	-
<i>Prepare and identify the BOM and its Specification for Solar PV Cell</i>	30	20	-	-
PC11. Discuss various Materials used in solar cell manufacturing	10	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Silicon Wafer, Chemicals, PPEs, Carrier & Trolley, Gases, Quartz Boat, Graphite Boat Spare, Tweezer, Tube cleaning rod, Printing Screen, Squeegee, Paste, Squeegee holder, Paper Roll, etc.	10	-	-	-
PC13. Technical Specifications & uses, role /importance in cell performance.	10	-	-	-
PC14. Demonstrate and show each component and explain their application in Solar cell	-	10	-	-
PC15. Prepare the BOM of Solar cell	-	10	-	-
NOS Total	60	40	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N3105
NOS Name	Describe the basics of PV Cell, prepare and identify the bill of material and its specification
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Process Line Operator
NSQF Level	4
Credits	2
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	24/02/2029
NSQC Clearance Date	25/02/2026

Qualification Pack

SGJ/N3104: Perform Solar cell manufacturing- Chemical Process

Description

This NOS covers the knowledge and practical skills required to perform chemical processes in solar cell manufacturing, including use of tools, etching techniques, process control, and defect identification in PV cells.

Scope

The scope covers the following :

- Covers chemical processes, the role of chemicals, and wetline techniques used in solar cell manufacturing.

Elements and Performance Criteria

Solar Cell Manufacturing - Chemical Process

To be competent, the user/individual on the job must be able to:

- PC1.** Explain Chemical process general requirement and types
- PC2.** Discuss tools used for Chemical process
- PC3.** Discuss purposes of Chemical Process
- PC4.** Explain Operation of the tools
- PC5.** Discuss how to Avoid wrong practices during Chemical Process.
- PC6.** Discuss the Statistical Process Control
- PC7.** Explain Identification of Defects & their criticality in PV Cell.
- PC8.** Demonstrate how to perform etching.
- PC9.** Demonstrate the bath change, make-up. techniques with hands on experience
- PC10.** Demonstrate use of Chemical tools during the process
- PC11.** Show how to check quality defect in products.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Chemical process requirements, types, and purposes involved in solar cell manufacturing.
- KU2.** Tools and equipment used in chemical processes and their proper operation.
- KU3.** Safe practices and precautions to avoid wrong practices during chemical processing.
- KU4.** Application of Statistical Process Control (SPC) to maintain process efficiency and quality.
- KU5.** Identification of defects and quality checking in PV cells during the chemical processing stage.

Generic Skills (GS)

User/individual on the job needs to know how to:

Qualification Pack

- GS1.** Follow safety procedures while handling chemicals and equipment.
- GS2.** Maintain cleanliness and proper chemical handling practices in the work area.
- GS3.** Use tools and equipment correctly during the chemical process.
- GS4.** Identify and report defects in PV cells during production.
- GS5.** Follow standard operating procedures during manufacturing processes.
- GS6.** Maintain accuracy while performing etching and chemical treatments.
- GS7.** Record and monitor process parameters for quality control.
- GS8.** Work effectively with the team during production activities.
- GS9.** Ensure proper handling, storage, and disposal of chemicals.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Solar Cell Manufacturing – Chemical Process</i>	24	26	-	-
PC1. Explain Chemical process general requirement and types	2	-	-	-
PC2. Discuss tools used for Chemical process	3	-	-	-
PC3. Discuss purposes of Chemical Process	5	-	-	-
PC4. Explain Operation of the tools	5	-	-	-
PC5. Discuss how to Avoid wrong practices during Chemical Process.	4	-	-	-
PC6. Discuss the Statistical Process Control	3	-	-	-
PC7. Explain Identification of Defects & their criticality in PV Cell.	2	-	-	-
PC8. Demonstrate how to perform etching.	-	10	-	-
PC9. Demonstrate the bath change, make-up. techniques with hands on experience	-	5	-	-
PC10. Demonstrate use of Chemical tools during the process	-	5	-	-
PC11. Show how to check quality defect in products.	-	6	-	-
NOS Total	24	26	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N3104
NOS Name	Perform Solar cell manufacturing- Chemical Process
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Process Line Operator
NSQF Level	4
Credits	1
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	24/02/2029
NSQC Clearance Date	25/02/2026

Qualification Pack

SGJ/N3103: Perform Solar cell manufacturing-Thermal Process

Description

This NOS covers the knowledge and practical skills required to perform thermal processes in solar cell manufacturing, including diffusion and PECVD operations, PN junction formation, ARC deposition, machine handling, and defect identification.

Scope

The scope covers the following :

- Diffusion and PECVD thermal processing in solar cell manufacturing.
- Equipment handling and defect inspection during the process.

Elements and Performance Criteria

Solar Cell Manufacturing - Thermal Process

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss the flow chart of Diffusion and PECVD manufacturing process.
- PC2.** Explain Machine Overview.
- PC3.** Discuss types of Chemical/Gas using for the process.
- PC4.** Explain bubbler change method.
- PC5.** Explain removal of defective broken cells from boat/tubes
- PC6.** Explain Solar cells PN Junction
- PC7.** Explain ARC Layer formation and deposition method
- PC8.** Discuss the Quality parameter and defects
- PC9.** Demonstrate solar cell PN Junction
- PC10.** Demonstrate ARC Thickness & Refractive Index
- PC11.** Demonstrate how to perform bubbler change.
- PC12.** Demonstrate graphite boat repair or assembly process.
- PC13.** Demonstrate to check defects.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Flow of diffusion and PECVD processes in solar cell manufacturing.
- KU2.** Machine overview and operation used in thermal processes.
- KU3.** Types of chemicals and gases used during the process.
- KU4.** PN junction formation and ARC layer deposition in solar cells.
- KU5.** Quality parameters, defect identification, and basic maintenance practices (bubbler change, boat handling).

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Operate diffusion and PECVD equipment following standard procedures.
- GS2.** Handle gases, bubblers, and process tools safely.
- GS3.** Perform basic maintenance such as bubbler change and graphite boat handling.
- GS4.** Inspect solar cells to identify defects and quality issues.
- GS5.** Follow safety, cleanliness, and workplace discipline during thermal processing.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Solar Cell Manufacturing – Thermal Process</i>	24	26	-	-
PC1. Discuss the flow chart of Diffusion and PECVD manufacturing process.	2	-	-	-
PC2. Explain Machine Overview.	2	-	-	-
PC3. Discuss types of Chemical/Gas using for the process.	1	-	-	-
PC4. Explain bubbler change method.	3	-	-	-
PC5. Explain removal of defective broken cells from boat/tubes	3	-	-	-
PC6. Explain Solar cells PN Junction	4	-	-	-
PC7. Explain ARC Layer formation and deposition method	4	-	-	-
PC8. Discuss the Quality parameter and defects	5	-	-	-
PC9. Demonstrate solar cell PN Junction	-	5	-	-
PC10. Demonstrate ARC Thickness & Refractive Index	-	5	-	-
PC11. Demonstrate how to perform bubbler change.	-	5	-	-
PC12. Demonstrate graphite boat repair or assembly process.	-	6	-	-
PC13. Demonstrate to check defects.	-	5	-	-
NOS Total	24	26	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N3103
NOS Name	Perform Solar cell manufacturing-Thermal Process
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Process Line Operator
NSQF Level	4
Credits	1
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	24/02/2029
NSQC Clearance Date	25/02/2026

Qualification Pack

SGJ/N3102: Perform Solar cell manufacturing- Metallization Process

Description

This NOS covers the metallization process in solar cell manufacturing, including screen printing, firing in furnace, visual inspection, defect identification, and performance testing using EL test and sun simulator.

Scope

The scope covers the following :

- Screen printing and firing process in solar cell metallization.
- Visual inspection, defect identification, and performance testing using EL test and sun simulator.

Elements and Performance Criteria

Solar Cell Manufacturing – Metallization Process

To be competent, the user/individual on the job must be able to:

- PC1.** The metallization process flow.
- PC2.** Machine Overview with operation control
- PC3.** Types of Chemicals in paste form uses
- PC4.** Discuss how to perform Screen printing
- PC5.** Explain Important checkpoints in process control
- PC6.** Discuss how to check its output current, voltage and power of solar cell
- PC7.** Explain Visual inspection to identify defects, Consequences of several process.
- PC8.** Explain how to perform EL test of solar cell and identify any dead and low power cells, short circuit cells, cracks, etc.
- PC9.** Explain use of sun simulator for performing IV test characteristics of solar module
- PC10.** Demonstrate screen mounting technique
- PC11.** Demonstrate how to operate Halm.
- PC12.** Demonstrate how to perform IV/ EL test
- PC13.** Demonstrate how to perform the visual Inspection
- PC14.** Demonstrate how to generate efficiency report

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Metallization process flow in solar cell manufacturing.
- KU2.** Screen printing process and machine operation.
- KU3.** Use of paste chemicals in metallization.
- KU4.** IV characteristics testing using sun simulator.
- KU5.** Visual inspection, EL testing, and defect identification in solar cells.



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Follow safety and workplace procedures during metallization operations.
- GS2.** Handle machines, tools, and materials carefully.
- GS3.** Maintain cleanliness and organization at the workstation.
- GS4.** Identify and report defects or abnormalities in solar cells
- GS5.** Communicate and coordinate effectively with the production team.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Solar Cell Manufacturing – Metallization Process</i>	24	26	-	-
PC1. The metallization process flow.	3	-	-	-
PC2. Machine Overview with operation control	3	-	-	-
PC3. Types of Chemicals in paste form uses	2	-	-	-
PC4. Discuss how to perform Screen printing	2	-	-	-
PC5. Explain Important checkpoints in process control	1	-	-	-
PC6. Discuss how to check its output current, voltage and power of solar cell	4	-	-	-
PC7. Explain Visual inspection to identify defects, Consequences of several process.	1	-	-	-
PC8. Explain how to perform EL test of solar cell and identify any dead and low power cells, short circuit cells, cracks, etc.	4	-	-	-
PC9. Explain use of sun simulator for performing IV test characteristics of solar module	4	-	-	-
PC10. Demonstrate screen mounting technique	-	6	-	-
PC11. Demonstrate how to operate Halm.	-	5	-	-
PC12. Demonstrate how to perform IV/ EL test	-	5	-	-
PC13. Demonstrate how to perform the visual Inspection	-	5	-	-
PC14. Demonstrate how to generate efficiency report	-	5	-	-
NOS Total	24	26	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N3102
NOS Name	Perform Solar cell manufacturing- Metallization Process
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Process Line Operator
NSQF Level	4
Credits	1
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	24/02/2029
NSQC Clearance Date	25/02/2026

Qualification Pack

SGJ/N3101: Perform packing of solar PV Cell

Description

This NOS covers the process of packing solar PV cells, including visual inspection, use of appropriate packaging materials, proper stacking and handling, and safe transportation of packed cells.

Scope

The scope covers the following :

- Inspection and packing of solar PV cells using appropriate packaging materials.
- Proper stacking, handling, and transportation of packed cells.

Elements and Performance Criteria

Packing of Solar PV Cell

To be competent, the user/individual on the job must be able to:

- PC1.** Discuss how to visually inspect the completed cell for quality of materials and workmanship.
- PC2.** Explain how to check proper packaging material for cell.
- PC3.** Discuss how to pack cell in properly designed cartons for transportation
- PC4.** Discuss how to keep cell in to stacked in a box.
- PC5.** Discuss the uses of separators
- PC6.** Identify processes where material and resources utilization can be optimized.
- PC7.** Demonstrate the Packing process and handling of Solar cell
- PC8.** Demonstrate use of Fork lift machine for moving of solar cell pallet from one place to other.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Solar PV cell packing process and workflow.
- KU2.** Visual inspection of completed cells for material quality and workmanship
- KU3.** Selection and use of appropriate packaging materials and cartons
- KU4.** Proper stacking of cells in boxes and use of separators for protection.
- KU5.** Methods to optimize material and resource utilization during packing

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Handle solar PV cells carefully to avoid damage during packing.
- GS2.** Follow proper packing procedures and safety guidelines.
- GS3.** Maintain cleanliness and organization at the packing area.



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- GS4.** Operate material handling equipment such as forklifts safely.
- GS5.** Coordinate with team members to ensure efficient packing and movement of cells.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Packing of Solar PV Cell</i>	50	50	-	-
PC1. Discuss how to visually inspect the completed cell for quality of materials and workmanship.	10	-	-	-
PC2. Explain how to check proper packaging material for cell.	10	-	-	-
PC3. Discuss how to pack cell in properly designed cartons for transportation	10	-	-	-
PC4. Discuss how to keep cell in to stacked in a box.	10	-	-	-
PC5. Discuss the uses of separators	5	-	-	-
PC6. Identify processes where material and resources utilization can be optimized.	5	-	-	-
PC7. Demonstrate the Packing process and handling of Solar cell	-	25	-	-
PC8. Demonstrate use of Fork lift machine for moving of solar cell pallet from one place to other.	-	25	-	-
NOS Total	50	50	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N3101
NOS Name	Perform packing of solar PV Cell
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Process Line Operator
NSQF Level	4
Credits	2
Version	2.0
Last Reviewed Date	25/02/2026
Next Review Date	24/02/2029
NSQC Clearance Date	25/02/2026

Qualification Pack

SGJ/N0147: Maintain personal health & safety in a manufacturing facility

Description

This unit is about maintaining health & safety in a manufacturing facility by following established safe work procedure, maintaining personal protective equipment as well as proper use of required tools and equipment

Scope

The scope covers the following :

- Establish and Follow safe work procedure
- Use and maintain personal protective equipment
- Identify and mitigate safety hazards
- Demonstrate safe and proper use of required tools and equipment

Elements and Performance Criteria

Maintain personal health & safety in a manufacturing facility

To be competent, the user/individual on the job must be able to:

- PC1.** identify the requirements for safe work area.
- PC2.** discuss how to Administer first aid.
- PC3.** discuss how to identify the personal protective equipment used for the specific purpose.
- PC4.** discuss how to identify the electrical and chemical hazards associated with photovoltaic installations.
- PC5.** discuss how to identify work safety procedures and instructions to Handle Sharp and heavy component in a solar PV manufacturing.
- PC6.** discuss occupational health & safety standards and regulations for installation of Solar PV system.
- PC7.** show how to Administer first aid.
- PC8.** show how to Identify the electrical and chemical hazards associated with photovoltaic installations.
- PC9.** show how to Identify and perform work safety procedures and instructions to handle sharp and heavy component in a solar PV Manufacturing.
- PC10.** show how to good housekeeping practices and infection control guidelines.
- PC11.** show how to Dispose- off any waste materials in accordance with safe working practices and procedures.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** company's installation policy
- KU2.** company's customer support policy

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- KU3.** document information using appropriate corporate forms
- KU4.** obtain authorization from specified field safety officer and supervisor
- KU5.** relevant personal protective equipments required within the manufacturing facility
- KU6.** relevant standards and regulations to be followed in the manufacturing facility
- KU7.** occupational health and safety (OHS) standards to be followed in the manufacturing facility
- KU8.** risk identification and mitigation procedure for safe work in a manufacturing facility
- KU9.** knowhow of tools & tackles required to carry out the work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** fill up documentation applicable to one's role
- GS2.** read vernacular/english language
- GS3.** read and understand manuals, health and safety instructions, memos, other company documents
- GS4.** ability to read from different sources- books, screens in machines and signage
- GS5.** understand the various colour codes, as per standard electrical, mechanical and civil nomenclature
- GS6.** express statements or information clearly so that others can hear and understand
- GS7.** participate in and understand the main points of simple discussions
- GS8.** respond appropriately to any queries
- GS9.** communicate with peers, supervisor and sub-ordinates
- GS10.** follow organisation rule- based decision making process
- GS11.** take decision with systematic course of actions and/or response
- GS12.** plan and organize work schedule to meet deadlines
- GS13.** plan to utilise time and equipment's effectively
- GS14.** work constructively and collaboratively with others
- GS15.** follow organisational code of conduct
- GS16.** manage relationships with customers with intent on satisfying its requirements for service delivery
- GS17.** recognize problems and search for solutions
- GS18.** choose best methods to complete assigned tasks
- GS19.** approach relevant authority when required
- GS20.** apply domain knowledge, observations and data to select course of action to perform tasks related to the job role
- GS21.** critically evaluate information obtained from customers, supervisor and co-workers to perform day to day activities
- GS22.** ask questions for better understanding

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain personal health & safety in a manufacturing facility</i>	25	25	-	-
PC1. identify the requirements for safe work area.	3	-	-	-
PC2. discuss how to Administer first aid.	4	-	-	-
PC3. discuss how to identify the personal protective equipment used for the specific purpose.	4	-	-	-
PC4. discuss how to identify the electrical and chemical hazards associated with photovoltaic installations.	4	-	-	-
PC5. discuss how to identify work safety procedures and instructions to Handle Sharp and heavy component in a solar PV manufacturing.	5	-	-	-
PC6. discuss occupational health & safety standards and regulations for installation of Solar PV system.	5	-	-	-
PC7. show how to Administer first aid.	-	5	-	-
PC8. show how to Identify the electrical and chemical hazards associated with photovoltaic installations.	-	5	-	-
PC9. show how to Identify and perform work safety procedures and instructions to handle sharp and heavy component in a solar PV Manufacturing.	-	5	-	-
PC10. show how to good housekeeping practices and infection control guidelines.	-	5	-	-
PC11. show how to Dispose- off any waste materials in accordance with safe working practices and procedures.	-	5	-	-
NOS Total	25	25	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SGJ/N0147
NOS Name	Maintain personal health & safety in a manufacturing facility
Sector	Green Jobs
Sub-Sector	Renewable Energy
Occupation	Health & Safety
NSQF Level	4
Credits	1
Version	4.0
Last Reviewed Date	25/02/2026
Next Review Date	24/02/2029
NSQC Clearance Date	25/02/2026

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification.

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Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SGJ/N3105.Describe the basics of PV Cell, prepare and identify the bill of material and its specification	60	40	0	0	100	25
SGJ/N3104.Perform Solar cell manufacturing- Chemical Process	24	26	0	0	50	12
SGJ/N3103.Perform Solar cell manufacturing-Thermal Process	24	26	0	0	50	12
SGJ/N3102.Perform Solar cell manufacturing- Metallization Process	24	26	0	0	50	12
SGJ/N3101.Perform packing of solar PV Cell	50	50	0	0	100	13
SGJ/N0147.Maintain personal health & safety in a manufacturing facility	25	25	-	-	50	13
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	13
Total	227	223	-	-	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.